Morning Paper Report - Tuesday, July 14, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Identical Attentional Capture with Different Working Memory Representation Precision

Liangliang Yi, Ruikang Zhong, Haibo Zhou, Daoqun Ding, Yutong Liu, Xinxin Xiang, et al.
Behavioral Science, 2026

Source: Semantic Scholar; matched terms: working memory representation, visual search, working memory, precision, attentional guidance

Link: https://www.semanticscholar.org/paper/9eb015124d8a11775633902b313a6450ec16c60d

Goal: Attention can be automatically captured by the distractor that matches the representation of working memory (WM) in search tasks, impairing visual search efficiency and resulting in attentional capture effects

Method: Two experiments were conducted to combine the attentional capture paradigm with continuous delayed-estimation tasks

Results: Attention can be automatically captured by the distractor that matches the representation of working memory (WM) in search tasks, impairing visual search efficiency and resulting in attentional capture effects

Conclusion: The findings show that the size of attentional capture effects based on WM is unaffected by the distribution of WM resources Attentional capture effects may reflect the attention bias of WM representation that occurs in preparation stage of memory-based attentional guidance.

2. The locus of visual search's interference on visual working memory.

Zachary Hamblin-Frohman
Journal of Experimental Psychology: Human Perception and Performance, 2022

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: https://www.semanticscholar.org/paper/840cfacc7813fe14aebc3c82b118e78bca3a21f2

Goal: Visual attention and visual working memory (VWM) are interacting systems

Method: Over five experiments, different attentional tasks were completed in the retention period of a difficult change detection task

Results: It is proposed that interference arises from the attentional selective process of choosing the target item out of the nontargets (PsycInfo Database Record (c) 2022 APA, all rights reserved).

Conclusion: It is proposed that interference arises from the attentional selective process of choosing the target item out of the nontargets (PsycInfo Database Record (c) 2022 APA, all rights reserved).

3. Parietal-driven visual working memory representation in occipito-temporal cortex.

Yaoda Xu
Current Biology, 2023

Source: Semantic Scholar; matched terms: visual working memory, working memory representation, working memory

Link: https://www.semanticscholar.org/paper/565e1aa52cf942ae362e593afcbbbec7ea140ce1

Goal: Human fMRI studies have documented extensively that the content of visual working memory (VWM) can be reliably decoded from fMRI voxel response patterns during the delay period in both the occipito-temporal cortex (OTC), including early visual areas (EVC), and the posterior parietal cortex (PPC).1,2,3,4 Further work has revealed that VWM signal in OTC is largely sustained by feedback from associative areas such...

Method: Taking advantage of a recent finding showing that object representational geometry differs between OTC and PPC in perception,10 here we find that, during VWM delay, the object representational geometry in OTC becomes more aligned with that of PPC during perception than with itself during perception This finding supports the role of feedback in shaping the content of VWM in OTC, with the VWM content of OTC more...

Results: Human fMRI studies have documented extensively that the content of visual working memory (VWM) can be reliably decoded from fMRI voxel response patterns during the delay period in both the occipito-temporal cortex (OTC), including early visual areas (EVC), and the posterior parietal cortex (PPC).1,2,3,4 Further work has revealed that VWM signal in OTC is largely sustained by feedback from associative areas such...

Conclusion: This finding supports the role of feedback in shaping the content of VWM in OTC, with the VWM content of OTC more determined by information retained in PPC than by the sensory information initially encoded in OTC.

4. Modulation of similarity on the distraction resistance of visual working memory representation.

Yanliang Sun, Lixue Wang, Nana Sun, Shouxin Li
Psychophysiology, 2022

Source: Semantic Scholar; matched terms: visual working memory, working memory representation, working memory

Link: https://www.semanticscholar.org/paper/0f3e6f915ce6d7d743f0dee03cddedea158580b9

Goal: Controversy exists regarding the distraction resistance of priority items in visual working memory (VWM)

Method: We investigated the effect of similarity on distraction resistance of relevant (color) and irrelevant (shape) feature representations (Experiments 1-2), and the neural mechanism of this effect using event-related potentials (ERPs; Experiment 3)

Results: Behavioral results showed distractors impaired the accuracy of dissimilar items when relevant features were memorized and of similar items when irrelevant features were memorized under simultaneous presentation of similar and dissimilar items

Conclusion: Our results support the available resource threshold account, suggesting that VWM is a flexible and intelligent system despite its limited capacity.

5. A certain future strengthens the past: Knowing ahead how to act on an object prioritizes its visual working memory representation.

Caterina Trentin, G. Rinaldi, Magdalena Chorzepa, M. Imhof, H. Slagter, Christiann L. Olivers
Journal of Experimental Psychology. Learning, Memory and Cognition, 2024

Source: Semantic Scholar; matched terms: visual working memory, working memory representation, working memory

Link: https://www.semanticscholar.org/paper/dcf6ab5c6210d9eecc85485ce44539fbd8ca19e5

Goal: Findings from recent studies indicate that planning an action toward an object strengthens its visual working memory (VWM) representation, emphasizing the importance of sensorimotor links in VWM

Method: In three eye-tracking experiments, we asked participants to memorize a visual stimulus for a subsequent memory test, whereby they performed a specific hand movement toward memory-matching probes

Results: Across the three experiments, we found moderate evidence that knowing in advance how to act on an object prioritized its mnemonic representation, as expressed in an increased attentional bias toward it

Conclusion: Findings from recent studies indicate that planning an action toward an object strengthens its visual working memory (VWM) representation, emphasizing the importance of sensorimotor links in VWM